Project Description

This course project will give students the opportunity to learn about cool methods in machine learning; do research; and get your hands dirty with data. Please **work in groups of 3 students on the projects**.

Something **new** should be **learned** as a result of your project. Your project may innovate some new ML methodology, or make an improvement to an existing method. It is also ok to learn how to apply existing ML models in a new application area. However, it is not ok to swap out one datafile for another in an existing script, or to produce a study whose main contribution is making a call to a ".fit()" function from an existing library. (This project will be judged against a semester's worth of effort.) If your project is application based, it must: compare across multiple methods, hyperparameters, and ways of pre-processing data; make use of multiple datasets; interpret results and models; list best practices for future researchers in the area with empirical (and possibly theoretical) backing.

Deliverables

The main **deliverables** are the following:

1. A project write-up, 8 pages NeurIPS format. (Note this requires using LaTeX, the standard markup language for scientific and technical manuscripts.)
2. Open source repository with executable code for methods developed.
3. Spotlight Presentation.

Project Proposal

Each group must submit a *one* page (minimum 11 size font) summary of their proposed project. (Proposal does not have to be in NeurIPS format.) The proposal will be judged according to how well you address Heilmeier's criteria:

- **What** are you trying to do? Articulate your objectives using absolutely no jargon.
- **How** is it done today, and what are the limits of current practice?
- What's **new** in your approach and **why** do you think it will be successful?
- **Who** cares? If you're successful, what difference will it make?
- What are the **risks** and the **payoffs**?
- **How long** will it take *(make a timeline)* and what have you achieved so far?
- How will you **determine success**?

Particularly, the proposal should contain the following information:

- What project idea are you pursuing?
    - Clearly state the task that you are performing, and the goals of the methods you will explore (e.g. better accuracy, sharper samples, quicker computation, etc.). *What do you (and readers of your report) hope to learn from the project?*

- What exactly are the methods that you propose to explore?
  - This should include a plan of baseline simple methods, existing methods, and extensions/modifications.
- What datasets will you consider?
  - Have a plan for generating synthetic data where the ground truth is known.
  - What real-world datasets will you try? (Are there existing benchmarks?)
- What will your evaluation criteria be? (This may not be obvious.)
  - Have a quantitative score, even if it is only a proxy for a score that one would truly want.
  - If there are qualitative scores that you will consider, then describe a systematic, unbiased way that you will obtain them.
- What are the challenges and possible drawbacks of your proposed directions?
  - What are your plans for dealing with these?
- What framework/language will things be implemented in?
  - Is there an existing library that you will base your code on? (This is probably a good idea if possible.)

Submit your proposal as a pdf file**.** Include all names in the proposal text and file name.